

Redesigning Human-Robot Interaction in Response to Robot Failures: a Participatory Design Methodology

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ABSTRACT

People are increasingly encountering robots in public spaces. To increase the robustness of such in-the-wild robotic applications and to achieve their designed outcomes, existing research focuses on improving the technical reliability of robots and identifying effective strategies to prevent or recover from technical failures. However, in human-robot interaction (HRI), a user's perception of a robot failure may not necessarily relate to technical issues. We focus on understanding users' perception of robot behaviours and interactions within the context of a public space. In our exploratory study using a novel participatory design methodology, participants designed robot behaviours for applications in public spaces, and tested their design both in a simulator and on the physical robot. We investigate how participants' perception and expectations change during this iterative participatory prototyping process, especially when the robot exhibits erratic or unexpected behaviours. Our work provides insights on users' perception of robot failures, and how users' design of robot behaviours shifts as they observe the robot within the spatial and social context.

CCS CONCEPTS

• **Human-centered computing** → **Participatory design**; *Web-based interaction*; • **Applied computing** → *Sociology*.

KEYWORDS

Human-Robot Interaction, Social Robotics, Participatory Prototyping

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1 INTRODUCTION

We are witnessing an increasing presence of robots in today's public spaces, such as airports, hospitals, or cafes. The COVID-19 pandemic has accelerated the introduction of robotic technology and automation with the goal of enhancing human capabilities, improving amenity, or replacing people in tasks that are dull, dirty, or dangerous [30]. However, even the most advanced robot in a public space can have difficulties achieving its intended outcomes, including executing designed behaviours or appropriately interacting with people, especially in less structured environments.

Existing research on robot failures has focused on the technical aspects with a robot-centered perspective. When a robot is deployed in public spaces, people may interact with it either by design or unintentionally. Thus, it is important to understand how these potential interactors perceive a robot's behaviours, especially when erratic or unexpected robot behaviours occur. Our work follows an interdisciplinary and human-centered perspective to design human-robot interaction (HRI) systems in public spaces. We focus on understanding users' expectations, perceptions, and reception of robot behaviours when placed within public space contexts, and how witnessing unfamiliar, erratic or unexpected robot behaviours changes their design of HRI functions. To support this goal, we

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deploy a novel research methodology with an online format of conducting participatory design workshops with both simulated and physical robots, extended from our previous work [31]. In our exploratory study using such a participatory design workshop format, we identified emerging themes for users' perception of robot behaviours and how they address failure recovery or prevention through redesigning the robot behaviours or the HRI system. Our main research questions are:

- (1) What do users consider as robot failures?
- (2) What is the impact of robot failures on users' expectations and design of robot behaviours?
- (3) How does participatory design with a physical robot change users' expectations and design of robot behaviours compared to participatory design with a robot simulator?

Our work contributes to current HRI and design research by investigating empirical and conceptual understandings of robot behaviours and robot failures. Current understanding of robot "failure" is limited. Our study provides insights on how people respond to perceived robot failures, and identify effective strategies to manage, mitigate or recover from them. This is critical to advancing current robotic research and achieving the intended outcomes of a robotic system [32]. The proposed iterative design and evaluation methodology facilitates investigation and prevention of robot failures through design approaches.

2 RELATED WORK

As current HRI systems are far from being error-free, it is important to understand robot failures and to identify effective strategies for preventing or recovering from failures. A recent review by Honig and Oron-Gilad [13] has shown that most existing studies on robot failures focused on technical aspects such as software or hardware failures, while relatively fewer studies have investigated the interactive and human aspects. HRI studies that address failures from a human-centered perspective focus on how a user's trust towards a robot [34] or a user's verbal and non-verbal behaviours [10] are influenced by technical failures of the HRI system. However, it is unclear whether a user's perception of a robot failure always aligns with technical failures in the robot's functions. Our exploratory study provides an opportunity to investigate robot failures from the end-users' perspective, and to investigate how these users adjust their design of robot behaviours to address failures that they observe. This allows us to arrive at a user-centered definition of "robot failures".

In order to understand robot failures as perceived by the users, we adopted participatory design as our methodology. Participatory design is a research methodology that involves end-users when developing new technologies in order to incorporate their domain expertise and manage their expectations of the system [18], using participatory prototyping [19] and design ethnography [23]. Recent HRI studies have demonstrated the effectiveness of participatory design methodologies for designing and evaluating more nuanced and contextualized robot applications. For example, Winkle et al. [36] engaged therapists to develop robot behaviours that can increase patients' engagement in rehabilitative therapies. Similarly, Lee and Riek [17] conducted a participatory design workshop with researchers and older adult participants to investigate how existing

or new assistive robots can support successful aging. Participatory design has also been applied to HRI applications for mental well-being for teens [4] and educational robots for visually impaired children [20].

These existing HRI studies focus on one-on-one interaction scenarios with specific populations. In contrast, our project focuses on robots deployed in public spaces, with varied interaction goals, and a diverse user population. Understanding robots in the wild and in society is an emerging topic [33]. Recent HRI studies have begun to address robots in public spaces, for example, social robots for entertainment, advertisement and other services in shopping malls were studied [9, 26, 28]. However, these studies were often focused on functional aspects with a robot-centered view, such as identifying effective methods for detecting user engagement [3]. In this project, we maintain a user-centered perspective instead. The majority of participatory design studies in HRI focus on prototyping new robot functions. However, participants often do not get the opportunity to develop and test out their designed prototypes with an actual robot in context. This can lead to a lower acceptance of the designed robot functions, especially when erratic or unexpected robot behaviours occur that contradict the users' expectations [1]. Furthermore, potential users typically do not get to nominate what they themselves would prefer or expect a robot to do in a given setting. Our participatory design workshop methodology allows participants to design behaviours and then test these behaviours first in a simulated environment and then on a physical robot. After observing the execution, participants can make iterative changes to improve the robot behaviours design addressing their observations. This allows us to investigate users' perceptions and expectations of robots as they design new behaviours and observe realisations of those behaviours both in simulation and contextualized on the physical robot.

3 METHODOLOGY

Our goal is understanding expectation and perception of robot failures. To facilitate this goal, we developed a participatory design methodology consisting of small group design workshops engaging end-users in the design of robot applications for public spaces. The participants first attended a participatory design workshop with a robot simulator (Section 3.2). While a simulator allows for agile prototyping, it limits users' ability to observe and design interactive robot behaviours and contextualize the spatial and social factors in which these interactions take place. As these spatial contexts and interactive behaviours are key to understanding users' expectations of robots, we conducted a second workshop to include a physical robot (Section 3.3). COVID-19 physical distancing restrictions necessitated an online format for the workshop to ensure the safety of our participants and researchers. We therefore developed a novel methodology for conducting participatory design workshops with robots, utilizing live-streaming to provide the contextualization and spatialisation required for designing and evaluating interactive robot behaviours.

3.1 Participants and tools

We recruited 12 Monash University staff members (5 males, 7 females) from several faculties who had no prior experience with

robots or programming. This allowed us to have a participant population with sufficient variability [11]. We intend to further extend the diversity of participants in future work to be more representative of the general population, by including those outside of universities, who may have a less academic background. The participants attended the workshops remotely via a video conferencing platform (Zoom) and provided written consent for their participation and for recording of the sessions. One participant was unable to attend the second workshop with the physical robot due to technical difficulties, leaving 11 participants for the robot-based workshop. The study protocol was reviewed and approved by our institutional Human Research Ethics Committee.

We chose the Pepper robot as a representative of robots currently deployed in public spaces for conducting our workshops [22]. For the robot simulator and live monitoring of the physical robot's status, such as its camera feed, we used Pepper's accompanying simulator Choregraphe [25]. To support participants to develop behaviours for the robot, we used Robot Interface from Zero Expertise (RIZE) [8] as the end-user programming interface. RIZE is a modular and distributed end-user development framework that enables the creation of visual programming environments for robotics [7]. RIZE follows a component-based methodology, in which key robot functions, such as text-to-speech, are encapsulated as building blocks that a user can customize to some extent (e.g., specifying the sentences the robot will say). These individual building blocks are then combined through logic structures (e.g., if-then) or along a timeline to form a more complex robot behaviour. Participants were supported by a technical expert during the sessions, whom we refer to as the *translator*. The translator answered participants' questions about the robot and suggested possible implementations for robot behaviours proposed by the participants. Participants were also provided with a *cheat-sheet* list of all available robot capabilities in RIZE, to help navigate the various robot functions offered for the workshop.

3.2 Online participatory design workshop with a robot simulator

We first conducted a participatory design workshop in the simulator environment (Figure 1). This simulator-based workshop allowed the participants to get familiar with the programming interface and the robot's capabilities. It also facilitated investigation on the participants' initial expectations of the robot and the interaction context, i.e., the public space they had in mind when designing the robot's behaviours. During the simulator-based workshop, participants began with an open discussion on what they think is a robot and their prior experiences with robots in public spaces. This helped them imagine and develop specific behaviours for a Pepper robot in a public space by drawing on any similar interactions they may have already had [24]. This was followed by a participatory prototyping session, in which the participants were randomly divided into two groups with similar demographic, each designing a set of robot behaviours for a public space of their choice with the use of RIZE, the cheat-sheet, and the support of the translator. Group 1 envisioned Pepper at a shopping mall, while Group 2 envisioned Pepper at a tourist attraction. After the programming session, the two groups rejoined, demonstrated their designs of

robot behaviours and explained the rationales behind. Description and analysis of the simulator-based workshop is detailed in our previous work [31].

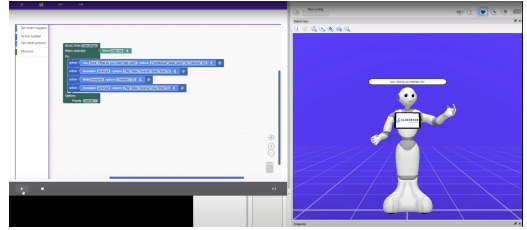


Figure 1: The RIZE end-user programming interface (on the left) and the robot simulator (on the right).

3.3 Online participatory design workshop with a physical robot

We next conducted a second workshop with a physical robot to include spatial and interactive context during the participatory design process. The two-hour workshop with the physical robot was conducted a month after the simulator-based workshop. The same participants attended both workshops, except for one participant who dropped out of the robot-based workshop due to connectivity issues. This allowed us to compare their perceptions and designs of robot behaviours across these two distinctive settings.

Before the robot-based workshop, the participants answered a pre-workshop questionnaire, which contained the GodSpeed questionnaire [2] for capturing their impression of the likeability and perceived intelligence of the robot, the Trust questionnaire [21] for capturing their trust towards the robot, and the Reliability questionnaire [29] for capturing the perceived reliability of the robot. They also provided an estimate of how successful they expected the robot to be at executing a designed behaviour and achieving its intended outcomes.

In the first 30 minutes of the robot-based workshop, participants were introduced to the physical robot. During the previous simulator-based workshop, both groups conceptualised one very similar robot behaviour: attending to a lost child. Thus, we chose this lost-child robot behaviour as the starting point for the robot-based workshop. Starting from this shared behaviour provides a common grounding from which the designs can be compared. In this lost-child behaviour, the robot identifies a child in distress based on the facial expression of sadness. It then approaches the child, performs a gesture of looking left and right, and asks the child “Hi there, are you okay? We can wait together until someone comes. Here, hold my hand if you want”. It then reaches its left hand out, opens and closes its left hand. Afterwards it asks “Do you want to see something cool? Please stand back a little”, and performs a dance that imitates an elephant. At the end, the robot says to the child “Someone is on their way”.¹ The participants watched a demonstration of the lost-child behaviour on the physical robot during the introduction.

¹Demonstration of the original lost-child robot behaviour, which was used as the starting point of the participatory design workshop using the physical Pepper robot: https://youtu.be/6Hn-34P_9Zg

After the introduction, participants engaged in a one-hour participatory prototyping session, in which they were randomly assigned into two independent groups. During this session, each group had a remote access to a Pepper robot and performed iterative changes to the lost-child behaviour. As shown in Figure 2, during each iteration, participants implemented changes to the robot behaviour in RIZE, then viewed the execution of the edited behaviour on the physical remote robot through a live-feed of the robot and through robot status monitoring in Choregraphe. After observing the execution, participants discussed their observations and proposed new edits to the design. Note that an experimenter acted as the lost child during this workshop for demonstrating and testing the robot behaviours. Following the participatory prototyping session, in the final 30 minutes of the robot-based workshop, the two groups rejoined for demonstration and discussion.

Participants answered a post-workshop questionnaire to further capture their experiences and reflections. This post-workshop questionnaire contains the GodSpeed questionnaire, Trust questionnaire, and Reliability questionnaire, as in the pre-workshop questionnaire. In addition, participants were asked to give an example of an unexpected event during the workshop. They also provided their perception of the robot's execution of the designed behaviour and the improvements needed for the robot to be more useful and acceptable in public spaces.

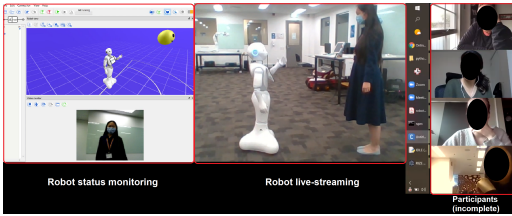


Figure 2: Online participatory design workshop with a physical robot.

4 OBSERVATIONS AND ANALYSES

We analyzed emerging themes in our workshop using inductive coding of the workshop recording. We studied how participants' perceptions and design of robot behaviours changed as a result of observing unexpected or erratic robot behaviours that occurred in physical context. Here we focus on participants' discussion during the workshop, especially their iterative design and testing of robot behaviours. Additional quantitative analysis will be reported in [6].

4.1 Unexpected or erratic robot behaviours

Examples of unexpected or erratic robot behaviours identified by the participants included sensory aspects that only physical encounters with the robot could reveal, such as loud motor noises or failures in recognizing faces or facial expressions. In general, there are two types of robot behaviours that the participants perceive as erratic or unexpected. One is technical failures in the execution of designed behaviours, such as when a voice trigger is not recognized, perhaps due to noise in the environment. The other is when a behaviour is correctly executed but does not meet expectations or

intended outcomes, or has an effect that the participant did not foresee. For example, one participant observed how a sentence spoken by the robot is perceived as inappropriate when placed in context: "One thing that struck me is the [robot's] comment that *someone is coming*. [sic] Actually when it's written down it looked fairly benign. But the robot says it you sort of wonder *Who's coming? Is he good or bad?* That's kind of creepy actually." While the first type of robot failure can be identified and addressed by robot-centered technical approaches, such as improving the robustness of the robot's speech recognizer, the second type of perceived robot failure can only be understood and addressed through user-centered and contextualized approaches. Therefore, it is critical for an HRI system to adapt to context and user's reception of its services.

When a robot failure occurred, participants discussed the potential causes. In these discussions, two major causes of robot failure are identified. The first cause is limitation in the robot's capabilities given the complexity of the task. As one participant stated: "Maybe it was too ambitious to jump straight into this [having the robot attend to a lost child]. I understand how relevant this application is, but maybe it's too much for what we have available now." The second cause is misalignment with context, such as characteristics of the user, the public space, or social context of the interaction. For example, a participant questioned whether it is appropriate for Pepper to approach a lost child at all, as "it doesn't have an identity badge, it doesn't wear a uniform, those were the things that you [as a child] recognize that someone is safe to talk to". Another participant discussed the possibility of encountering a child with autism, and how the robot's emotion recognition function may mis-classify this child's facial expression as sadness, which would trigger the lost-child behaviour by mistake. Observing and analyzing these robot failures brought the complexity of public spaces to the participants' attention. For example, one participant estimated that Pepper will not succeed in executing the lost-child behaviour, explaining: "As our second workshop showed, this behaviour requires a series of complicated and contextual interactions / decisions which are too difficult for Pepper to navigate, at least in any kind of reliable, safe, or sensitive way."

Participants' discussion of robot failures revealed that, particularly in unpredictable and dynamic settings, success or failure was not solely determined by the capabilities of the robot. That is, "failure" was not a deficit or malfunction, but rather a shortcoming of the robot's existing capabilities, the setting, the participants' pre-existing understanding of robots and their views on what would be a useful task for the robot to be able to do. Our study revealed that the success of robots, particularly in public space, is the result of an assemblage of related factors, so future studies must be carefully contextualised to best design robots for public spaces. In Sections 4.2 and 4.3, we will further discuss how robot failures influence the participants' design and perception of robot behaviours.

4.2 Changes in design of robot behaviours

As a result of these perceived robot failures, both groups made major changes to their design of the lost-child robot behaviour

during the workshop.² Interestingly, the two groups took opposite approaches in revising their designs.

Group 1, who envisioned Pepper at a shopping mall, focused on the robot's lack of ability to address nuances in context, such as correctly identifying a lost child. They decided to take away the agency of the robot and leave the task of deciding whether someone is lost to a human. Additionally, they extended the concept of "lost" to adults who need directions. Instead of approaching a lost person proactively, the robot serves as a passive helper. In the case of someone informing the robot there is a lost child, the robot will notify a staff member and stay with the child until the staff member arrives and takes over. In the case of an adult asking the robot for directions, it will lead the person to their destination. Group 1 explained their design decision as follows: "Just the sheer amount of social complexity involved in any of these kinds of decisions [identifying a lost person] and how much of these is really implicit in our decision-making that has to be explicit in Pepper's decision-making, [are the reasons] which limits Pepper's ability [to] be the active agent. Because there is just so much adversary involved in exercising that agency, which is why it [Pepper] works better as a more passive tool. It [Pepper] is more like a kiosk, rather than like a person". Here, "robot failure" was addressed by changing the robot's task, rather than the robot itself.

Group 2, who envisioned Pepper at a tourist attraction, focused on the robot's lack of responsiveness. They decided to add decision points in the interaction flow and included new branches addressing possible scenarios. In particular, after identifying a lost child, the robot will ask the child "do you want to see something cool?". If the answer is "yes", the robot will engage in a game interaction, in which it first performs the elephant dance, then asks the child to "be an elephant too"; if the answer is "no" or if the child does not respond, the robot will not engage in this game interaction and instead stay with the child. In all scenarios, similar to Group 1, the robot notifies a human staff member and stays with the child until the staff member arrives and takes over. Group 2 explained their design decision as follows: "The main thing we wanted to fix with Pepper was the sense that Pepper was just running through the motions with the child and there was no pause in between actions and really no interaction. We felt that because children can be very unpredictable, especially if they are in a situation of distress, we thought that it would be a good idea to make the engagement with Pepper more interactive to try and de-escalate any of the stress that the child's feeling. So what we did was we worked mainly with the elephant move and [...] hopefully engaging the child to actually interact with Pepper, and then an encouragement at the end saying *That's a good job being an elephant*. That was the main thing that we were focusing on and that was to try and include the child in the engagement situation. And also to increase the humanness of Pepper and guiding the child to act in a certain way because previously it felt like Pepper was running through the script". Group 2's approach to "robot failure" differed from Group 1, in that it modified the actions of the robot itself, rather than modifying the scenarios they imagined it in.

Both groups emphasised the differences between seeing only the simulator and seeing the actual robot, and how that motivated their revision in robot behaviour designs. Firstly, having the physical robot facilitates testing of interactive behaviours and brings sensory and embodied aspects to the fore that may not be as easily perceived in the simulator. As the participants identified, "It was clearer how difficult it is to capture the triggers. In the simulator we assumed that Pepper captured the sad expression and then in practice we see that it doesn't always happen that smoothly. When we have Pepper in the space and interacting with other people, it's where we have the possibility of seeing things going wrong. Because in the simulator, if we code it right it won't go wrong, but in the real life setting it can go wrong." Being able to investigate the reliability of robot functions led the participants to revise several robot functions. An example is how Group 1 replaced voice trigger with touch trigger, as they stated "we actually and pretty explicitly in the beginning did not want touch to be a trigger. We were like *I don't think we should want people touching Pepper. It would be much better to have a vocal trigger*. But that wasn't really working so it forced us into touch for the sake of having a reliable behaviour". Secondly, the placement of the actual robot in a physical space allowed participants to situate robot's behaviours in context. As the participants suggested, "It even comes down to just how loud the motors were the first time it moved, that's very abrupt in a sonic way. It was just all that detail that you were abstracting away in the computer simulation that became what was more important than what we thought would be, like how loud it is, how abrupt it is, how it moves, things like that". These physical robot behaviours did not exhibit technical failures but were perceived as erratic or unexpected by the participants. Again, this builds a picture of "robot failure" as highly contingent, with implications for future HRI studies.

4.3 Changes in perception and expectation of the robot

The participants reflected on their workshop experience mainly in terms of how their expectations of the robot have been grounded, and how they understand the limits of the technology better. As one participant stated, "I think we were moderately successful [in implementing the desired robot behaviours], but really only because we kept making adjustments to account for what Pepper could do – if not all the time, then at least reliably. This meant lowering our expectations and changing how we understood Pepper as an object to be programmed". The participants also discussed the fact that Pepper being a humanoid robot can raise users' expectations of its capability in performing social interaction: "Pepper looks kind of human and we were putting this effort to making the interaction as closest [sic] as possible to human interactions. But the more we make Pepper human-like, the bigger the expectation that people have that it will actually act like a human. We are trying to make Pepper feel like a natural and social interactions but this only raises the bar of what people would expect Pepper to do." Such expectation discrepancy, which can lead to disappointment or misplaced trust, remains an open challenge in HRI [27]. Our participatory design methodology provides a possible approach to address expectation

²Revised lost-child robot behaviour (demonstrated in simulator) designed by Group 1 (<https://youtu.be/7scuBNN-zU8>) and by Group 2 (<https://youtu.be/P8l7HOytXBs>) in the participatory design workshop using the physical robot.

discrepancy: it grounds users' expectations closer to the robot's actual capabilities by exposing users to robot failures.

To address technical limitations of the robot, a suggestion that the participants agreed upon is to “start with something simple, something easier or more predictable”. They explained: “through these two workshops we realized how complex the situation actually is, how many variables are involved, and how many ethical issues are even more important because of the child. So maybe it was too ambitious to jump straight into this [lost-child behaviour]. Maybe starting with smaller and more simple situations / scenarios will give insights to then move to a more complex scenario”. This adjustment of expectation of the robot's capability led the participants to change the role they imagined the robot to serve in a public space. As one participant suggested, “I imagine people would see it [Pepper] as a novelty and interact with it for fun. I'm not sure that people would seek genuine information or assistance from the robot. I think for some period of time there would [be] mild fascination, particularly with children, but I suspect the interest would fade relatively quickly over time. Much of this has to do with its gimmicky presentation.” Such loss of interest over long-term use remains a major challenge in HRI [12], and the key for a robot to remain engaging after the novelty effect wears off is to improve its context-awareness and adaptivity [32]. This was also identified by the participants as “Robots will need to be more adaptable to situations [to make it more useful in public spaces]”.

5 DISCUSSION

RQ1: What do users consider as robot failures?

Our exploratory study suggests that whether a robot behaviour is perceived as a failure is a complex and contingent combination of factors beyond the robot behaviour itself. This means that technical failures of the robot's functions are only one part of how behaviours are understood and assessed by people who encounter robots. However, the majority of existing HRI studies have focused on a robot-centered perspective for addressing robot failures [13]. Beyond advancing functional aspects of robot designs, in order to identify and reduce robot failures during in-the-wild HRI, it is equally important to address the social-emotional aspects of robot designs. This facilitates reducing social errors in HRI [32], such as the inappropriate speech response example “someone is coming” given in Section 4.1. Thus, it is crucial to adopt a much more expansive frame for analysis that incorporates robot behaviours, users' existing views and dynamic and growing understandings of those behaviours, changing interaction contexts and the sensory aspects of engaging with a robot to perform a holistic evaluation of HRI systems. This follows but also extends, for example, the USUS evaluation frameworks that combines the usability, social acceptance, user experience, and societal impact aspects of HRI [35].

RQ2: What is the impact of robot failures?

Participants reported that their expectations were brought closer to the robot's actual capabilities after the iterative prototyping session. Users' realization that robots are fallible extends beyond the participatory design workshop. For example, when asked how successful they expect the robot to be at providing services in a public space, one participant pointed out that “the voice command recognition wasn't perfect during the workshop, so it's likely to fail in a loud

space”. Expectation grounding is especially important when people are likely to over-estimate a technology. In the case of a robot, overly optimistic expectations are common, due to the influence of science fiction and mass media [14]. However, this case study also showed how flexible these expectations are, and how they change over even a brief workshop session, or when the setting changes from a simulator to a physical robot. An interesting approach taken by the participants to address robot failures is to take away part of the automation and leave some decisions to humans, such as having a person notify the robot there is a lost child instead of having the robot identifying a lost child itself. This sheds light on how robotic technologies and automation can be adopted to effectively enhance human capabilities through human-robot collaboration [5]. It also shows how perception of robot behaviours can readily move on from a perception of “failure” not by changing the robot, but through changes to expectations. The proposed methodology can be applied to facilitate such human-robot collaboration by helping users to build and modify their understanding of their robot teammate, while enabling iterative development of robot functions that are acceptable and effective for achieving the collaborative goals of the human-robot team, even as these goals remain flexible.

RQ3: Participatory design with a physical robot vs. with a robot simulator?

Our exploratory study supports the effectiveness of using a physical robot for incorporating additional context compared to using a simulator, particularly in studies of interaction. Details that are often missing in a simulator, such as spatialisation or the sound of the robot's movement or voice, are found to contribute to users' perception of robots significantly in our study and in other HRI studies [15, 16]. However, robot simulators allow agile prototyping, which is also beneficial for iterative participatory design. Thus, both simulator-based and robot-based participatory design can yield benefits for understanding users' perception of robot behaviours.

6 CONCLUSIONS AND FUTURE DIRECTIONS

We proposed a novel methodology for conducting online participatory design workshops with robots. Observations made during our exploratory study provide us with a first indication of the efficacy of such workshops in analyzing users' perception of robots and HRI applications situated in the physical and interactive context. We were able to capture changes in users' perception of a robot's capabilities as well as in their design of the robot's behaviours and roles, resulting from observing robot failures. Our analyses of what the participants consider to be unexpected or erratic robot behaviours have shown that the perception of a robot's failures extends beyond technical failures – and moreover we were able to move towards an understanding of “failure” as being located in the contingent and emplaced encounters between humans and robots. A robot behaviour that executed successfully can still be perceived by the users as a failure because of its misalignment with the interaction context, although these understandings can change readily. Moreover, for an HRI system to achieve its intended outcome and to be perceived as socially acceptable, the ability to adapt to variances in the context is key. However, such adaptivity is missing in current HRI systems. Our study contributes to the understanding of users' expectations and perception of robots in

public spaces, as well as how designs of robot behaviours are influenced by limitations in a robot's capabilities. Beyond the context of robots in public space, the proposed remote workshop format can be applied to other HRI participatory design practices, particularly when there are impediments to face-to-face studies. The proposed participatory design workshop can facilitate the design and evaluation of robot behaviours with other application goals, such as education or mental well-being.

One limitation of our current workshop methodology is that it relies on retention of the same participant group across the simulator-based and robot-based workshops. Therefore, we will investigate a compressed format of the workshop that combines the simulator-based and robot-based approaches within one session. Furthermore, we plan to organize in-person workshops, in which we will investigate how users' design and perceptions change when they share the same space with the robot and have direct interactions with it. Moreover, when it is safe to do so, we aim to place robots in actual public spaces and recruit occupants of the space to gather insights embedded in the spatial and social contexts inherent to those spaces. Another limitation of our exploratory study is that a single robot platform was used. However, various robots with different functionality and appearance designs are deployed in public spaces. As Pepper represents a class of robots that are humanoid and designed with a focus on social interaction, we plan to conduct studies with robots that have a non-humanoid appearance or are designed with a focus on physical functionalities, such as cleaning. This will allow us to further investigate how characteristics of a robot influence users' expectations and perceptions. In addition, we plan to expand the robot functions implemented in the RIZE end-user programming interface to offer increased flexibility in the users' design choices.

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REFERENCES

- [1] Shiri Azenkot, Catherine Feng, and Maya Cakmak. 2016. Enabling building service robots to guide blind people a participatory design approach. In *HRI'16*. IEEE, 3–10.
- [2] Christoph Bartneck, Dana Kulić, Elizabeth Croft, and Susana Zoghbi. 2009. Measurement instruments for the anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety of robots. *International journal of social robotics* 1, 1 (2009), 71–81.
- [3] Atef Ben-Youssef, Giovanna Varni, Slim Essid, and Chloé Clavel. 2019. On-the-Fly Detection of User Engagement Decrease in Spontaneous Human-Robot Interaction Using Recurrent and Deep Neural Networks. *International Journal of Social Robotics* 11, 5 (2019), 815–828.
- [4] Elin A Björling, Emma Rose, Andrew Davidson, Rachel Ren, and Dorothy Wong. 2019. Can we keep him forever? Teens' engagement and desire for emotional connection with a social robot. *International Journal of Social Robotics* (2019), 1–13.
- [5] Christina Bröhl, Jochen Nelles, Christopher Brandl, Alexander Mertens, and Verena Nitsch. 2019. Human-robot collaboration acceptance model: development and comparison for Germany, Japan, China and the USA. *International Journal of Social Robotics* 11, 5 (2019), 709–726.
- [6] Pamela Carreno-Medrano, Leimin Tian, Aimee Allen, Shanti Sumartojo, Michael Mintrom, Enrique Coronado, Gentiane Venture, Elizabeth Croft, and Dana Kulić. 2021. Iterative Changes of Robot Behavior Designs and Perceptions in End-User Programming. In *ICRA'21 (under review)*. IEEE.
- [7] Enrique Coronado, Fulvio Mastrogiovanni, Bipin Indurkha, and Gentiane Venture. 2020. Visual Programming Environments for End-User Development of Intelligent and Social Robots, a Systematic Review. *Journal of Computer Languages* (2020), 100970.
- [8] Enrique Coronado, Fulvio Mastrogiovanni, and Gentiane Venture. 2018. Development of intelligent behaviors for social robots via user-friendly and modular programming tools. In *ARSO'18*. IEEE, 62–68.
- [9] Mary Ellen Foster, Bart Craenen, Amol Deshmukh, Oliver Lemon, Emanuele Bastianelli, Christian Dondrup, Ioannis Papaioannou, Andrea Vanzo, Jean-Marc Odobez, Olivier Canévet, et al. 2019. MuMMER: Socially Intelligent Human-Robot Interaction in Public Spaces. *arXiv preprint arXiv:1909.06749* (2019).
- [10] Manuel Giuliani, Nicole Mirnig, Gerald Stollnberger, Susanne Stadler, Roland Buchner, and Manfred Tscheligi. 2015. Systematic analysis of video data from different human-robot interaction studies: A categorization of social signals during error situations. *Frontiers in psychology* 6 (2015), 931.
- [11] Greg Guest, Arwen Bunce, and Laura Johnson. 2006. How many interviews are enough? An experiment with data saturation and variability. *Field methods* 18, 1 (2006), 59–82.
- [12] Guy Hoffman. 2019. Anki, Jibo, and Kuri: What We Can Learn from Social Robots That Didn't Make It. *IEEE Spectrum* (2019).
- [13] Shanee Sarah Honig and Tal Oron-Gilad. 2018. Understanding and Resolving Failures in Human-Robot Interaction: Literature Review and Model Development. *Frontiers in psychology* 9 (2018), 861.
- [14] Aike C Horstmann and Nicole C Krämer. 2019. Great expectations? Relation of previous experiences with social robots in real life or in the media and expectancies based on qualitative and quantitative assessment. *Frontiers in psychology* 10 (2019), 939.
- [15] Takamune Izui and Gentiane Venture. 2019. Correlation Analysis for Predictive Models of Robot User's Impression: A Study on Visual Medium and Mechanical Noise. *International Journal of Social Robotics* (2019), 1–15.
- [16] Adrian Benigno Latupeirissa, Panariello Claudio, and Roberto Bresin. 2020. Exploring emotion perception in sonic HRI. In *Sound and Music Computing Conference, Torino, 24-26 June 2020*. Zenodo, 434–441.
- [17] Hee Rin Lee and Laurel D Riek. 2018. Reframing assistive robots to promote successful aging. *ACM Trans. on HRI* 7, 1 (2018), 1–23.
- [18] Hee Rin Lee, Selma Šabanović, Wan-Ling Chang, Shinichi Nagata, Jennifer Piatt, Casey Bennett, and David Hakken. 2017. Steps toward participatory design of social robots: mutual learning with older adults with depression. In *HRI'17*. 244–253.
- [19] Noortje Marres, Michael Guggenheim, and Alex Wilkie. 2018. *Inventing the social*. Mattering Press.
- [20] Oussama Metatla, Sandra Bardot, Clare Cullen, Marcos Serrano, and Christophe Jouffrais. 2020. Robots for Inclusive Play: Co-designing an Educational Game With Visually Impaired and sighted Children. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [21] Bonnie M Muir. 1994. Trust in automation: Part I. Theoretical issues in the study of trust and human intervention in automated systems. *Ergonomics* 37, 11 (1994), 1905–1922.
- [22] Amit Kumar Pandey and Rodolphe Gelin. 2018. A mass-produced sociable humanoid robot: Pepper: The first machine of its kind. *IEEE Robotics & Automation Magazine* 25, 3 (2018), 40–48.
- [23] Sarah Pink, Camille Santiago Caminha, Angelica Porto C De Souza, Renata Zilse, and Alex Sandro Gomes. 2017. Design Ethnography for Screenless Interaction Style: hands-on and no-hands in early morning routines. In *Proceedings of the XVI Brazilian Symposium on Human Factors in Computing Systems*. 1–10.
- [24] Sarah Pink, Vaïke Fors, and Mareike Glöss. 2018. The contingent futures of the mobile present: automation as possibility. *Mobilities* 13, 5 (2018), 615–631.
- [25] Emmanuel Pot, Jérôme Monceaux, Rodolphe Gelin, and Bruno Maisonnier. 2009. Choregraphe: a graphical tool for humanoid robot programming. In *RO-MAN'09*. IEEE, 46–51.
- [26] Alessandra Maria Sabelli and Takayuki Kanda. 2016. Robovie as a mascot: a qualitative study for long-term presence of robots in a shopping mall. *International Journal of Social Robotics* 8, 2 (2016), 211–221.
- [27] Lena T Schramm, Derek Dufault, and James E Young. 2020. Warning: This Robot is Not What it Seems! Exploring Expectation Discrepancy Resulting from Robot Design. In *Companion of the 2020 ACM/IEEE International Conference on Human-Robot Interaction*. 439–441.
- [28] Chao Shi, Satoru Satake, Takayuki Kanda, and Hiroshi Ishiguro. 2018. A robot that distributes flyers to pedestrians in a shopping mall. *International Journal of Social Robotics* 10, 4 (2018), 421–437.
- [29] Candace L Sidnera, Christopher Leea, Cory D Kidd, Neal Lesha, and Charles Richa. 2005. Explorations in engagement for humans and robots. *Artificial Intelligence* 166 (2005), 140–164.
- [30] Shanti Sumartojo, Daniele Lugli, Dana Kulić, Leimin Tian, Pamela Carreno-Medrano, Michael Mintrom, and Aimee Allen. 2020. *Robotic Logics of Public Space in the COVID Pandemic*. Retrieved Sep 17, 2020 from <https://www.>

- mediapolisjournal.com/2020/08/robotic-logics-of-public-space/
- [31] Leimin Tian, Pamela Carreno-Medrano, Shanti Sumartojo, Michael Mintrom, Enrique Coronado, Gentiane Venture, and Dana Kulić. 2020. User Expectations of Robots in Public Spaces: A Co-design Methodology. In *International Conference on Social Robotics*. Springer, 259–270.
 - [32] Leimin Tian and Sharon Oviatt. 2020. A Taxonomy of Social Errors in Human-Robot Interaction. *ACM Transactions on Human-Robot Interaction (to appear)* (2020).
 - [33] Ilaria Tiddi, Emanuele Bastianelli, Enrico Daga, Mathieu d'Aquin, and Enrico Motta. 2020. Robot-city interaction: Mapping the research landscape—a survey of the interactions between robots and modern cities. *International Journal of Social Robotics* 12, 2 (2020), 299–324.
 - [34] Suzanne Tolmeijer, Astrid Weiss, Marc Hanheide, Felix Lindner, Thomas M Powers, Clare Dixon, and Myrthe L Tielman. 2020. Taxonomy of Trust-Relevant Failures and Mitigation Strategies. In *Proceedings of the 2020 ACM/IEEE International Conference on Human-Robot Interaction*. 3–12.
 - [35] Astrid Weiss, Regina Bernhaupt, and Manfred Tscheligi. 2011. The USUS evaluation framework for user-centered HRI. *New Frontiers in Human-Robot Interaction* 2 (2011), 89–110.
 - [36] Katie Winkle, Praminda Caleb-Solly, Ailie Turton, and Paul Bremner. 2019. Mutual shaping in the design of socially assistive robots: A case study on social robots for therapy. *International Journal of Social Robotics* (2019), 1–20.